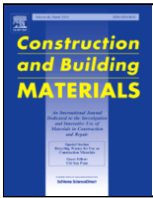




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Life cycle assessment of asphalt and cement pavements: Comparative cases in Shanxi Province

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ABSTRACT

In order to understand the energy consumption and environmental pollution level of Shanxi highway construction, the life cycle environmental and energy impact assessment of asphalt mixture and cement concrete were conducted. One representative cement concrete pavement and two asphalt mixture highways were selected for quantitative evaluation and LCA analysis. The total energy consumption values of Taichang highway are 20% higher than that of the Dayun highway. The global warming potential indicators of Taichang (asphalt mixture), Dayun (asphalt mixture), and Deda (cement concrete) are 181 kg eq CO₂, 285 kg eq CO₂ and 330 kg eq CO₂ respectively during the materials production stage. The results indicated that the environmental effects of global warming had the biggest impact on the life cycle of the asphalt mixtures and the respective highways. Based on the study results, environmental pollution was highest on the Taichang highway whilst the Deda highway had the highest values of freshwater toxicity. Furthermore, energy consumption during the construction stage and environmental impacts at the end-of-life stage were found to be alarmingly high. Thus, reducing the energy consumption during the material preparation and highway construction stages as well as controlling the environmental pollutants and emissions during service are key to conserving energy, preserving the surrounding environment, and optimizing the life cycle of Shanxi highways.

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1. Introduction

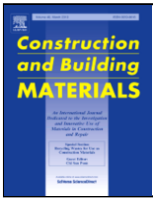
With the economic development, the transportation development of Shanxi Province has met a tremendous development opportunity. As of 2020, Shanxi Province had a total of 30 major highways. Taichang (Taiyuan to Changzhi and Jincheng), Dayun (Taiyuan to Datong and Yuncheng) and Deda (Deshengkou to Datong city) are the three principal highways of Shanxi Province. Due to the high energy consumption and construction/maintenance costs associated with these highways, it is necessary to comparatively evaluate and quantify their sustainability (both economically and environmentally) through conducting a life cycle assessment (LCA) [1–2].

LCA can be used for evaluating potential environmental pollutants and energy consumption throughout a product or structure's life [3–4].

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沥青与水泥路面生命周期评价：山西省案例比较

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摘 要

为了解山西省公路建设的能源消耗和环境污染水平，对沥青混合料和水泥混凝土进行了生命周期环境与健康影响评估。选取了一条代表性水泥混凝土路面和两条沥青混合料公路进行定量评价和LCA分析。太常公路的总能耗比大运公路高出20%。在材料生产阶段，太常（沥青混合料）、大运（沥青混合料）和德大（水泥混凝土）的全球变暖潜势指标分别为181 kg CO₂当量、285 kg CO₂当量和330 kg CO₂当量。结果表明，全球变暖的环境效应对沥青混合料及其所在公路的生命周期影响最大。根据研究结果，太常公路的环境污染最为严重，而德大公路的淡水毒性值最高。此外，施工阶段的能耗和退役阶段的环境影响被发现高得惊人。因此，减少材料制备和公路施工阶段的能源消耗，以及控制服役期间的环境污染物和排放，是节约能源、保护周边环境和优化山西省公路生命周期的关键。

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1. 引言

随着经济发展，山西省的交通发展迎来了巨大机遇。截至2020年，山西省共有30条主要公路。太昌（太原至长治和晋城）、大运（太原至大同和运城）和德大（得胜口至大同市）是山西省的三条主干公路。由于这些公路的高能耗及建设/维护成本，有必要通过进行生命周期评估（LCA）来比较和量化其可持续性（无论是经济方面还是环境方面）[1–2]。

LCA可用于评估产品或结构在整个生命周期中的潜在环境污染物和能源消耗。生命周期[3–4]。

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近期研究表明，沥青路面使用阶段的能耗受多种因素影响，包括材料类型、功能层结构和路面厚度[5–6]。除了相关的成本和高资源消耗潜力外，能耗还会导致严重的温室气体（CO₂）排放。因此，减少交通领域的CO₂排放已成为许多国家缓解国家碳排放的关键措施之一[7]。

中国已开展大量研究，采用生命周期评价（LCA）方法评估沥青路面环境影响[8]。研究表明，与再生沥青路面相比，橡胶改性沥青路面具有更高的环境影响[9]。同样，AzariJafari等人[10]研究了再生沥青路面的人体毒性潜力、金属释放及危险废物，发现底渣中迁移的污染物（Cd、Cr、Se和Ag）导致非饱和区污染水平极低。“从摇篮到坟墓”的全生命周期评价通常涉及多个阶段和活动，导致不同的能源与环境范式[11–12]。沥青路面

ment preservation such as use of thin overlays is not only cost-effective, but also brings significant environmental benefits [13–14]. Likewise, use of bio-asphalt materials could significantly reduce the energy consumption factor and environmental pollutants [15]. The literature shows that CO₂ emissions and environmental impacts related to the roads and highways are just one magnitude lower than that of bridges and tunnels [16]. Asphalt pavement can affect significantly the environment and economy of China [17–19]. Although there are many researches about LCA of asphalt pavement, there are lack of LCA of cement concrete pavement, and LCA analysis method of Shanxi highways based on the various stages of energy and environmental assessment are not clear.

Based on the above background, as a step towards minimizing energy consumption and reducing the environmental pollution, the LCA of representative highways in Shanxi Province, were comparatively evaluated in this study, with a focus on quantifying the energy and environmental impacts. The functional units, boundary conditions, life cycle inventory and data acquisition of LCA of Shanxi highway were proposed. LCA energy consumption evaluation included impacts of different highways and pavement types and impacts of different highway life cycle stages were investigated. Further, a life-cycle analysis method was also formulated based on the various stages of energy and environmental assessment. This study proposed the four environmental impact indicators (Global warming potential, Acidification potential, Freshwater toxicity, and Human toxicity) and its detailed evaluation compounds, which were used to verify the environmental impact indicators whether were suitable with the typical highways in Shanxi Province.

2. Study methodology

The LCA in this study was conducted in accordance with the ISO standard 14040/14044 [20] using the 2019 Chinese energy data. Three major highways, namely Taichang, Deda, and Dayun, respectively, in Shanxi Province of China were selected and used as the case study. The respective highway construction data were obtained from Shanxi Transportation Holdings Group CO., LTD.

As discussed in the subsequent text, the study methodology incorporated defining the scope of work and the corresponding research tasks. The study functional units, boundary conditions, and highway pavement structures are also discussed in this section of the paper.

2.1. Scope of work and research tasks

LCA is a useful tool for evaluating the energy consumption and the impacts of environmental pollution. Within this framework, the scope of work and research tasks for this study involved addressing the following objectives:

- Energy consumption assessment, mainly focusing gasoline and diesel usage;
- Evaluation of the environmental impacts of each stage of the LCA which included CO₂, CO, CH₄, SO₂, NO_x, NMVOC and PM2.5 emissions; and
- Identification of the potential environmental improvements by analyzing different types of asphalt mixture pavements and cement concrete pavements from a life cycle point of view.

As previously mentioned, the selection of the highway types to be studied was based on considering that there are three major highways in Shanxi Province (China), namely:

- Taichang (asphalt mixtures pavement),
- Deda (cement concrete pavement) and
- Dayun (asphalt mixtures pavement)

2.2. Study functional units and boundary conditions

The selected study functional units, which represents the fundamental unit throughout the whole LCA study, were each 1.0 km length of a highway pavement section. The pavement layer thicknesses for the different highways were different and were evaluated as such. The study included all the stages and activities that encompass materials production, transportation, construction, operations and maintenance work, and end-of-life during the life cycle process as per the “cradle-to-grave” method [21]. The “cradle-to-grace” method is that everything ranges from raw material acquisition to production, use and disposal.

The total pavement width for each of the three selected highways (namely Taichang, Deda, and Dayun), incorporating all the lanes, was about 27.5 m. The corresponding pavement structures, mostly comprising of the surface course, binder course, unbound base course, and sub-grade course, are shown in Table 1.

Other highway structures such as embankments, drainage systems and road markings were not included in this study because their contributions is negligible for highway construction and LCA [22].

The pavement surface layer thickness of the Taichang, Deda, and Dayun highways are 17 cm, 28 cm, and 16 cm, respectively. The boundary conditions and the flow sequence adopted for the study are shown in Fig. 1.

From Fig. 1, the flow sequence for LCA and the boundary conditions can be summarized as bullet-listed below:

- Materials production including raw materials and energy supplements.
- Transportation including raw materials transportation, earthwork removal/transportation and equipment transportation.
- Construction, which is made up of the pavement and base, or sub-base construction and includes all the stages of the road construction.
- Operations and maintenance work including maintenance, demdition, and reconstruction of the pavement, in order to ensure functionality, in terms of the bearing capacity, surface layer or thin overlay over the lifespan of the pavement construction.
- End-of-life, which consists of recycling, reuse, or disposal of solid waste, or regenerated material.

To sustain and maintain the average life span of road pavements, there is often a need (including preventive maintenance) to ensure suitable levels of service. The fundamental data of road pavement showed as Table 2 and these data obtained from Chinese Reference Life Cycle Database (CLCD) [23]. The database is that a national background LCI database consists of ca. 600 LCI datasets for key materials and chemicals, energy carriers, transport, and waste management. In this study, the useful lifetime of Taichang and Dayun highway pavements (asphalt) was assumed to be 15 years while that of Deda highway pave-

Table 1
Highway pavement structures.

Structure layer	Taichang	Deda	Dayun
Material type			
Upper layer	4 cm SMA-13 (asphalt)	28 cm cement concrete	4 cm SMA-13 (asphalt)
Middle layer	6 cm AC-20	–	5 cm AC-16
Lower layer	7 cm AC-25	20 cm cement stabilized macadam	7 cm AC-20
Base	33 cm cement stabilized macadam	36 cm ash stabilized soil	20 cm cement stabilized macadam
Sub-base	20 cm composite stabilized soil	20 cm gravel	20 cm cement stabilized macadam

Legend: AC = asphalt concrete, SMA = stone matrix (mastic) asphalt.

路面保存措施如使用薄层罩面不仅成本效益高, 而且带来显著的环境效益 [13–14]。同样, 使用生物沥青材料可显著降低能耗因子和环境污染物 [15]。文献表明, 道路和高速公路的二氧化碳排放及环境影响仅比桥梁和隧道低一个数量级 [16]。沥青路面能显著影响中国的环境和经济 [17–19]。尽管关于沥青路面生命周期评价的研究很多, 但缺乏水泥混凝土路面的生命周期评价, 且基于各阶段能源与环境评估的山西高速公路生命周期评价分析方法尚不明确。

基于上述背景, 为减少能源消耗和环境污染, 本研究对山西省代表性高速公路进行了生命周期评价对比评估, 重点量化能源与环境影响。提出了山西省高速公路生命周期评价的功能单位、边界条件、生命周期清单及数据采集方法。生命周期评价能耗评估包括不同高速公路和路面类型的影响, 以及不同生命周期阶段的影响。此外, 还基于各阶段能源与环境评估制定了生命周期分析方法。本研究提出了四个环境影响指标 (全球变暖潜力、酸化潜力、淡水毒性和人体毒性) 及其详细评估化合物, 用于验证这些环境影响指标是否适用于山西省典型高速公路。

2. 研究方法

本研究中的LCA遵循ISO标准14040/14044 [20], 采用2019年中国能源数据。选取了山西省境内的三条主要高速公路 (太昌、德大和大运) 作为案例研究。相应的公路建设数据来自山西交通控股集团有限公司。

如下文所述, 研究方法包括确定工作范围和相应的研究任务。本文本节还讨论了研究的功能单位、边界条件和公路路面结构。

2.1. 工作范围与研究任务

LCA是评估能源消耗和环境污染影响的有用工具。在此框架内, 本研究的工作范围和研究任务涉及解决以下目标:

- 能源消耗评估, 主要关注汽油和柴油的使用;
- LCA各阶段的环境影响评估, 包括CO₂、CO、CH₄、SO₂、NO_x、NMVOC和PM2.5排放;
- 通过从生命周期角度分析不同类型的沥青混合料路面和水泥混凝土路面, 识别潜在的环境改进。

如前所述, 研究公路类型的选择基于山西省 (中国) 有三条主要公路的考虑, 即:

- 太昌 (沥青混合料路面),
- 德大 (水泥混凝土路面),
- 大运 (沥青混合料路面)

2.2. 研究功能单元与边界条件

选定的研究功能单位代表整个生命周期评价研究中的基本单位, 每条高速公路路面段落长度为1.0公里。不同高速公路的路面层厚度各不相同, 并据此进行评估。研究涵盖了从“摇篮到坟墓”方法 [21] 的生命周期过程中涉及材料生产、运输、施工、运营与维护工作以及寿命终结的所有阶段和活动。“摇篮到坟墓”方法是指从原材料获取到生产、使用和处置的全过程。

所选三条公路 (即太昌、德大和大运) 的总路面宽度 (包括所有车道) 约为27.5米。相应的路面结构 (主要包括面层、粘结层、无结合料基层和路基) 如表1所示。

其他公路结构如路堤、排水系统和道路标线未纳入本研究, 因为它们对公路建设和LCA [22] 的贡献可忽略不计。

太昌、德大和大运高速公路的路面面层厚度分别为17厘米、28厘米和16厘米。研究采用的边界条件和流程序列如图1所示。

从图1可以看出, LCA的流程顺序和边界条件可归纳如下:

- 材料生产, 包括原材料和能源补充。
- 运输, 包括原材料运输、土方清除/运输和设备运输。
- 施工, 由路面和基层或底基层施工组成, 包括道路施工的所有阶段。

- 运营和维护工作, 包括路面的维护、拆除和重建, 以确保其功能性, 即在路面结构寿命周期内的承载能力、面层或薄层加铺。
- 寿命终止, 包括固体废物的回收、再利用或处置, 或再生材料。

为维持和保持道路路面的平均使用寿命, 通常需要进行养护 (包括预防性养护) 以确保适当的服务水平。道路路面的基本数据见表2, 这些数据来自中国生命周期参考数据库 (CLCD) [23]。该数据库是一个国家背景LCI数据库, 包含约600个关键材料与化学品、能源载体、运输和废物管理的LCI数据集。在本研究中, 太昌和大运高速公路路面 (沥青) 的使用寿命假定为15年, 而德大高速公路路

表1公路路面结构。

结构层	太昌	德大	大运
上层	材料类型 4 cm SMA-13 (沥青)	28厘米水泥混凝土	4厘米SMA-13 (沥青)
中层	6厘米AC-20	–	5 cm AC-16
下层	7 cm AC-25	20 cm 水泥 稳定碎石	7 cm AC-20
Base	33 cm 水泥 稳定碎石	36 cm 灰土稳定 soil	20 cm 水泥 稳定碎石
Sub-base	20厘米复合 稳定土	20厘米碎石	20厘米水泥 稳定碎石

图例: AC = 沥青混凝土, SMA = 沥青玛蹄脂碎石。

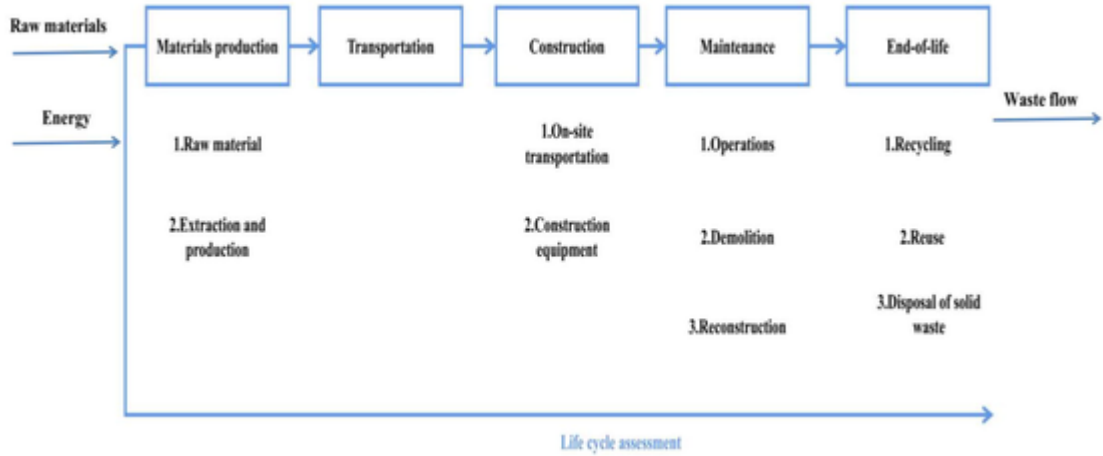


Fig. 1. Life cycle stages diagram

Table 2 The fundamental data of road pavement		
Data of raw materials		
Raw materials	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
Base asphalt	2830.760	1891.20
SBS modified asphalt	5412.220	2320.40
Rubber	3590	0970
Aggregate	31990	2380.163
Cement	340000	79850
Data of materials production (mixtures production)		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	33670	28260
AC13	33490	27810
AC16	32580	28050
AC20	33490	27810
AC25	307800	28260
Data of Transportation stage		
Transportation method	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
30-ton heavy diesel truck	0804	0075
Data of construction stage (Paving process of mixtures)		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	15860	1180
AC13	13220	0980
AC16	11580	0750
AC20	8810	0660
AC25	7340	0530
Data of construction stage (Rolling process of mixtures)		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	18600	1380
AC13	13230	0980
AC16	12850	0960
AC20	12480	0930
AC25	10390	0770
Data of maintenance stage		
Mixture type	Energy consumption (MJ/t)	Environmental pollutants (kg/t)
SMA-13	18290	30630
AC13	15970	26750
AC16	15250	25830
AC20	14800	24780
AC25	7970	11930

ment (concrete) was assumed to be 20 years. Maintenance stage, maintenance or preventive maintenance of the top layer of the pavement structure included an ultra-thin wearing course or skid-resistant surface layer, which was assumed to occur at the third of the asphalt pavements' service life (i.e., per 5 years/maintenance).

3. LCA formulation and data assembly

The LCA formulation including the software method used and the environmental impact indicators considered in the study are discussed in this section of the paper. The life cycle inventory and data acquisition for LCA are also discussed in this section.

3.1. LCA method used and environmental impact indicators

Life cycle assessments were computed using the materials balance method whilst the life cycle environmental impacts indicators were formulated using the ReCipe method and OpenLCA LCIA methods 1.5.7. In this study, the LCA was evaluated using the openLCA1.10.3 software [24]. Life cycle energy consumption and emission data were used based on the database of CML 2015. The fundamental energy consumption lists included carbon emission values (g/MJ), specific energy consumption (Q_{net} , MJ/kg), and emission factors (g/t). The eco-indicator for climate change was CO₂ emission while the eco-indicator for air acidification was SO₂.

The main eco-indicator of human toxicity were NO_x, CO, NMVOC, PM_{2.5}, NH₃ and heavy metals which can be represented by 1,4-dichlorobenzene equivalent. The eco-indicator for freshwater toxicity was hydrocarbon and heavy metal which could be represented by 1,4-dichlorobenzene equivalent. The eco-indicator for terrestrial ecotoxicity was part of NMVOC, hydrocarbon and heavy metals, which could be represented by 1,4-dichlorobenzene equivalent. In this study, the energy and environmental indicators or categories were evaluated and showed as follows:

- Global energy consumption (MJ);
- Global warming potential (GWP, kg eq CO₂);
- Acidification potential (AP, kg eq SO₂);
- Freshwater toxicity (FT, kg eq 1,4-dichlorobenzene included hydrocarbon and heavy metals); and
- Human toxicity (HT, kg eq 1,4-dichlorobenzene included NMVOC + PM_{2.5} + NO_x).

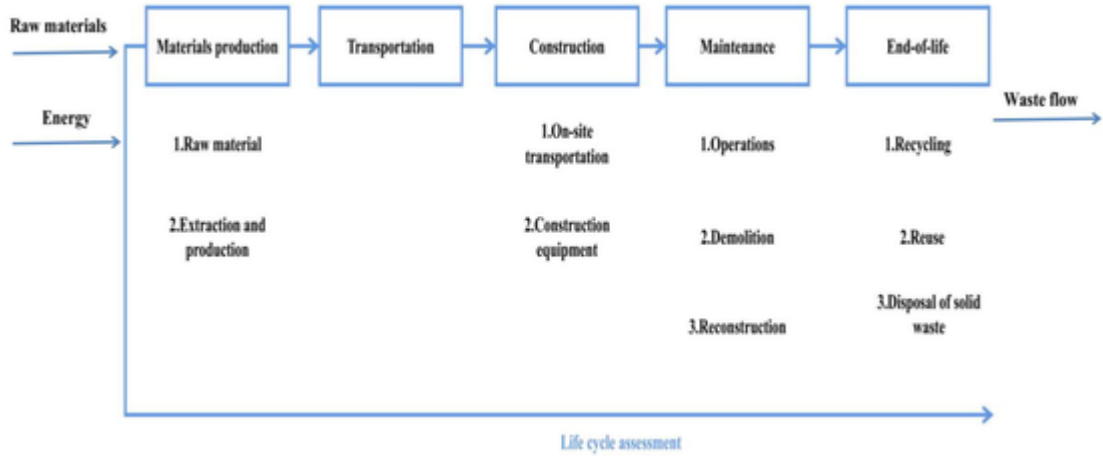


图1. 生命周期阶段示意图。

表2道路路面的基本数据。

原材料数据		
原材料	能耗 (MJ/t)	环境污染物 (kg/t)
基质沥青	2830.760	1891.20
SBS改性沥青	5412.220	2320.40
橡胶	3590	0970
骨料	31990	2.380*10 ⁻³
水泥	340000	79850
材料生产数据（混合料生产）		
混合料类型	能源消耗 (MJ/t)	环境污染物 (kg/t)
SMA-13	33670	28260
AC-13	33490	27810
AC-16	32580	28050
AC-20	33490	27810
AC-25	307.800	28260
运输阶段数据		
运输方法	能耗 (MJ/t)	环境污染物 (kg/t)
30吨重柴油卡车	0804	0075
施工阶段数据（混合料摊铺过程）		
混合料类型	能耗 (MJ/t)	环境污染物 (kg/t)
SMA-13	15.860 1.180	0980
AC-13	13220	0750
AC-16	11580	0660
AC-20	8810	0660
AC-25	7.340 0.550	
施工阶段数据（混合料碾压过程）		
混合料类型	能源消耗 (MJ/t)	环境污染物 (kg/t)
SMA-13	18600	1380
AC-13	13230	0980
AC-16	12850	0960
AC-20	12480	0930
AC-25	10390	0770
维护阶段数据		
混合料类型	能耗 (MJ/t)	环境污染物 (kg/t)
SMA-13	18290	30630
AC-13	15970	26750
AC-16	15250	25830
AC-20	14800	24780
AC-25	7970	11930

面（混凝土）的使用寿命假定为20年。在养护阶段，路面结构表层的养护或预防性养护包括超薄磨耗层或抗滑表层，假定在沥青路面使用寿命的三分之一时进行（即每5年养护一次）。

3. LCA 方法与数据收集

本文本部分讨论了 LCA 方法，包括所用软件方法以及研究中考虑的环境影响指标。本节还讨论了生命周期清单及 LCA 数据获取。

3.1. LCA 方法及环境影响指标

生命周期评估采用物料平衡法计算，而生命周期环境影响指标则采用 ReCipe 方法和 OpenLCA LCIA 方法 1.5.7 进行制定。本研究中，LCA 使用 openLCA1.10.3 软件进行评估 [24]。生命周期能耗和排放数据基于 CML 2015 数据库。基础能耗清单包括碳排放值 (g/MJ)、比能耗 (Q_{net} , MJ/kg) 和排放因子 (g/t)。气候变化生态指标为 CO₂ 排放，空气酸化生态指标为 SO₂。

人类毒性的主要生态指标包括 NO_x、CO、NMVOC、PM_{2.5}、NH₃ 和重金属，这些可用 1,4-二氯苯当量表示。淡水毒性的生态指标是碳氢化合物和重金属，可用 1,4-二氯苯当量表示。陆地生态毒性的生态指标是部分 NMVOC、碳氢化合物和重金属，可用 1,4-二氯苯当量表示。在本研究中，能源和环境指标或类别经过评估并如下所示：

- 全球能耗 (MJ)；
- 全球变暖潜能 (GWP, kg eq CO₂)；
- 酸化潜能 (AP, kg eq SO₂)；
- 淡水毒性 (FT, kg eq 1,4-二氯苯, 含碳氢化合物和重金属)；
- 人体毒性 (HT, kg eq 1,4-二氯苯, 含 NMVOC + PM_{2.5} + NO_x)

3.2. Life cycle inventory and data acquisition

Life cycle inventory (LCI) was conducted to quantitatively evaluate the energy and environmental inputs and outputs of the LCA system. The LCI data is shown in [Table 3](#).

The specific energy consumption of natural gas, crude oil, coal, gasoline, and uranium are 48.0 MJ/kg, 42.3 MJ/kg, 26.7 MJ/kg, 45.3 MJ/kg, and 8.5*10⁴ MJ/kg, respectively. The specific gravity of the raw material for (1 t) asphalt production were 20.1 kg/t (natural gas), 40.9 kg/t (crude oil), 1.03 kg/t (coal), 25.8 kg/t (gasoline), and 0.0001 kg/t (uranium), respectively. The energy consumption was calculated as the cumulative energy demand (MJ). The inventory data for the raw materials were obtained from the Ecoinvent database (Version 3.6).

Asphalt mixture or cement concrete pavement process is not only energy energy-consumption, but also includes environmental pollutants emissions such as CO₂, CO, CH₄, SO₂, NO_x, NMVOC and PM2.5. The LCI of environmental pollutants are showed in [Table 4](#).

The emission factors of the environmental pollutants listed in [Table 4](#) are very important for evaluating the LCA of the pavement. The total energy consumption of the raw materials for all the stages of the LCA are shown in [Table 5](#).

The different highways had different data of raw material consumption. These data were the basis for calculating the total energy consumption and environmental pollutants. The fundamental data for energy requirements of the machinery during mixture production, construction (mixing), and transposition were all obtained from the standard of JTGT 3832-2018 “Highway Engineering Budget Quota”.

The fuel consumption quota of the construction machinery of 1 work day is shown in [Table 6](#). Using this table, the total fuel consumption and energy consumption of the construction machinery were calculated. Thereafter, the environmental pollutants arising from construction machinery were estimated.

Table 3
LCI data for raw materials and energy.

Energy	Gas (MJ/kg)	Asphalt (kg/t)	Cement (kg/t)
Natural gas	480	201	–
Crude oil	423	409	–
Coal	267	103	960
Gasoline	453	258	–
Uranium	85*10 ⁴	0.0001	0.0005
Eurobitume	3227*10 ³	–	–
Total energy consumption of asphalt production	4900*10 ³	–	–

Table 4
LCI of emissions factors for the environmental pollutants

Emission factor	Asphalt (g/t)	Cement (g/t)
CO ₂	174244	605000
SO ₂	78	13
NO _x	70	16
NMVOC	33	36
PM2.5	16	95

Table 5
Fuel consumption quota of the construction machinery for 1 work day

Type	Taichang	Dayun	Deda
Crude oil (kg)	12608	14515	1558
Gasoline	1865	1626	2525
Diesel oil	5761	4878	12168
Electricity (kWh)	88902	78646	912566

Table 6
Total energy consumption of the raw materials and machine values

Type	Taichang	Deda	Dayun
Workday (d)	16230	25212	2525
Final aggregate asphalt mixture (t)	36000	0	32500
Cement concrete (t)	0	45068	0
Asphalt (t)	34860	0	3651
Fuel (t)	2924	0	3868
Aggregate clips (t)	15932	18215	12223
Crushed aggregate (t)	33152	62548	32515
Wheel loader (d)	1977	2515	2102
Mixing equipment (d)	740	625	780
Paving equipment (d)	879	548	915
Road roller (d)	5007	1246	5102
Dumper truck (d)	799	622	812
Sprayer truck (d)	123	1284	125

4. Results and discussions

The LCA results are presented, analyzed, synthesized, and discussed in this section of the paper. These results and analyses include energy consumption and environmental impact evaluations.

4.1. LCA energy consumption evaluation

The energy consumption evaluation included LCA for different highways and life cycle stages. These results for these LCA evaluations with respect to energy consumption are presented and discussed in this subsection of the paper.

4.1.1. Impacts of different highways and pavement types

The life cycle energy consumption impacts of different pavement types and different stages on emission were analyzed and compared to the traditional pavements. Reducing energy consumption and decreasing environmental pollution constitute some of the key measures to ensure sustainable development [25]. The energy consumption of the highways is shown in [Fig. 2](#).

As shown in [Fig. 2\(a\)](#), the total energy consumption of the Deda highway is smaller than that of Taichang highway and Dayun highways. These results demonstrate that Deda highway, with a cement concrete pavement structure had the smallest energy consumption and that cement concrete can be rated as a low energy consumption pavement technology when compared to asphalt.

Furthermore, the total energy consumption values of Taichang highway is greater than that of Dayun highway. According to the highways construction experts, Taichang highway and Dayun highway are comprised of almost similar pavement structures, particularly the upper surfacing layer - see [Table 1](#). Therefore, they should ideally have similar total energy consumption. However, the results ([Fig. 2\(a\)](#)) show that the total energy consumption values of Taichang highway are 20 % higher than that of the Dayun highway. As evident in [Table 1](#), the main reason for this is that the middle and base layer thickness of the Taichang highway are greater than that of Dayun highway, leading to an increase in then energy consumption.

According to the literature, the total average energy consumption of Chinese highways is 152524.42 MJ [26]. Compared to this study, the highways in Shanxi Province consume a lot of energy. The main reason is the unprofessional supervision [27]. To mitigate this situation, the government can use a public tender and impose energy consumption policies/regulations on designers, builders, inspectors, and clients. Unified training for the construction team can also be mandated.

4.1.2. Impacts of different highway life cycle stages

The life cycle was divided into five stages that included materials production, transportation, construction, operations/maintenance work, and end-of-life. As shown in [Fig. 2\(b\)](#), the energy consumption

3.2. 生命周期清单与数据获取

生命周期清单（LCI）旨在定量评估LCA系统的能源和环境输入与输出。LCI数据如[表3](#)所示。

天然气、原油、煤炭、汽油和铀的比能耗分别为48.0 MJ/kg、42.3 MJ/kg、26.7 MJ/kg、45.3 MJ/kg和8.5×10^{−4} MJ/kg。每生产1吨沥青的原材料消耗分别为：天然气20.1 kg/t、原油40.9 kg/t、煤炭1.03 kg/t、汽油25.8 kg/t、铀0.0001 kg/t。能耗以累计能源需求（MJ）计算。原材料清单数据来源于Ecoinvent数据库（3.6版本）。

沥青混合料或水泥混凝土路面施工过程不仅消耗能源，还排放CO₂、CO、CH₄、SO₂、NO_x、NMVOC和PM2.5等环境污染物。环境污染物生命周期清单见[表4](#)。

[表4](#)所列**环境污染物的排放因子对于评估路面LCA非常重要。各阶段原材料的总能耗如表5**所示。

不同高速公路的原材料消耗数据不同。这些数据是计算总能耗和环境污染物的基础。混合料生产、施工（拌和）及运输过程中机械所需的能源基础数据均来自标准JTGT 3832–2018《公路工程预算定额》。

1个工作日的施工机械燃油消耗定额见表6。利用该表计算了施工机械的总燃油消耗和能源消耗，进而估算了施工机械产生的环境污染物。

表3原材料与能源的LCI数据。

能源	Gas (MJ/千克)	沥青 (kg/t)	水泥 (kg/t)
天然气	480	201	–
原油	423	409	–
Coal	267	103	960
汽油	453	258	–
铀	8.5*10 ^{−4}	0.0001	0.0005
欧标沥青	3227*10 ^{−3}	–	–
沥青生产总能耗	4900*10 ^{−3}	–	–
生产			

表 4环境污染物排放因子的 LCI

排放因子	沥青 (g/t)	水泥 (g/t)
CO ₂	174,244	605,000
SO ₂	78	13
NO _x	70	16
NMVOC	33	36
PM2.5	16	95

表5施工机械每日工作1天的燃油消耗定额

Type	太昌	大运	Deda
原油（千克）	12608	14515	1558
汽油	1865	1626	2525
柴油	5761	4878	12168
电力（千瓦时）	88902	78646	912566

表6原材料与机械总能耗值

Type	太长	Deda	大运
工作日（天）	16230	25212	2525
细集料沥青混合料（吨）	36000	0	32500
水泥混凝土（吨）	0	45068	0
沥青（吨）	34860	0	3651
填料（吨）	2924	0	3868
骨料石屑（吨）	15932	18215	12223
破碎骨料（吨）	33152	62548	32515
轮式装载机（天）	1977	2515	2102
拌合设备（天）	740	625	780
摊铺设备（天）	879	548	915
压路机（天）	5007	1246	5102
自卸卡车（柴油）	799	622	812
喷洒车（柴油）	123	1284	125

4. 结果与讨论

LCA结果在本节中进行了呈现、分析、综合和讨论。这些结果和分析包括能源消耗和环境影响评估。

4.1. 生命周期评价能耗评估

能耗评估包括对不同高速公路及生命周期阶段的LCA分析。这些关于能耗的LCA评估结果在本小节中进行展示和讨论。

4.1.1. 不同高速公路与路面类型的影响

对不同路面类型和不同阶段的寿命周期能源消耗影响进行了分析，并与传统路面进行了比较。减少能源消耗和降低环境污染是确保可持续发展的关键措施 [25]。高速公路的能源消耗情况如[图2](#)所示。

如[图2\(a\)](#)所示，德大高速公路的总能源消耗小于太昌高速公路和大运高速公路。这些结果表明，采用水泥混凝土路面结构的德大高速公路能源消耗最小，与沥青相比，水泥混凝土可被视为低能耗路面技术。

此外，太昌高速公路的总能源消耗值大于大运高速公路。据高速公路建设专家介绍，太昌高速公路和大运高速公路的路面结构几乎相似，特别是上面层——参见[表1](#)。因此，理论上它们应具有相似的总能源消耗。然而，结果（[图2\(a\)](#)）显示，太昌高速公路的总能源消耗值比大运高速公路高20%。如[表1](#)所示，主要原因是太昌高速公路的中层和基层厚度大于大运高速公路，导致能源消耗增加。

根据文献，中国高速公路的总平均能耗为152524.42 MJ [26]。与此研究相比，山西省的高速公路能耗较高。主要原因在于监管不专业 [27]。为缓解这种情况，政府可以采用公开招标，并对设计者、建设者、检查者和客户实施能耗政策/法规。还可以强制要求对施工团队进行统一培训。

4.1.2. 不同高速公路生命周期阶段的影响

生命周期分为五个阶段，包括材料生产、运输、施工、运营/维护和寿命终止。如[图2\(b\)](#)所示，材料生产的能耗

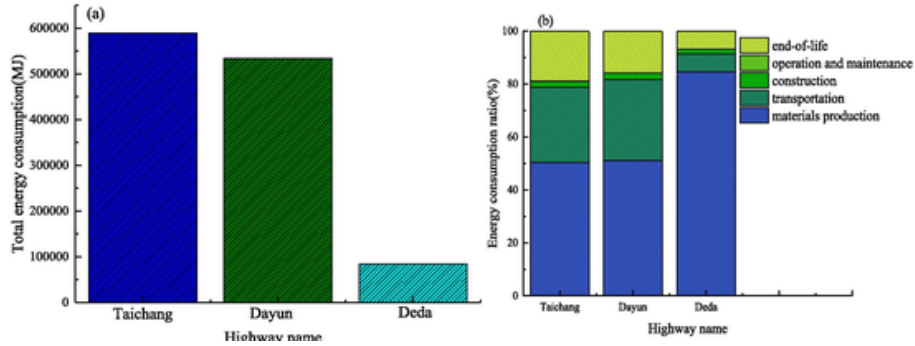


Fig. 2. Energy consumption (a) highway total energy consumption, (b) energy consumption at different life cycle stages

for materials production is the largest whilst the operation and maintenance stage has the smallest energy consumption values. Due to low energy consumption values, the numerical column for the operation and maintenance stage in Fig. 2(b) almost invisible.

From Fig. 2(b), it can also be observed that the combined energy consumption ratio for materials production and transportation stages was above 70 % of the whole life cycle of the highways. This illustrates that transportation and material production stages are the primary energy consumers during the life cycle of these three Shanxi highways. Compared with energy consumption of Inner Mongolia highways [28], the primary energy consumers stages are different during the life cycle. However, the energy consumption ratios for the different stages were different in the three highways. Compared to the energy consumption of the Deda highway with respect to materials production, the energy consumption values of other two highways (i.e., Dayun highway and Changda highway) were small. However, the energy consumption with respect to the construction stage seem to be similar for all the highways.

Overall, the results in Fig. 2(b) indicate that the energy consumption of cement concrete-based highways is significantly higher than that of asphalt mixtures-based highways in terms of materials production. Furthermore, it can also be concluded that materials production is the main energy consumption stage in the cement concrete-based highways, such as Deda, followed by transportation and end-of-life stages.

Likewise, materials production was similarly found to be the largest energy consumer in the asphalt-based highways (but numerically lower than the concrete-based highways) consecutively followed by the transportation and end-of-life stages. For all the three highways, Fig. 2(b) shows similar energy consumption for construction and negligible (i.e., less than 1.00 %) for the end-of-life stage. In comparison to asphalt-based highways, materials production and end-of-life stages were associated with the highest and least energy consumption, respectively, for concrete-based highways.

4.2. LCA environmental impacts evaluation

As discussed in the subsequent text, the environmental impacts evaluation included LCA for global warming, acidification potential, freshwater toxicity, and human toxicity.

4.2.1. Global warming potential (GWP) impacts

The factors responsible for global warming include truck, raw materials production, and paver emissions (CO₂) during the highway construction process. Minimizing CO₂ emissions is thus one of the key remedies to reduce global warming potential. The global warming potential (GWP) results are shown in Fig. 3(a).

As shown in Fig. 3(a), the global warming potential indicators of Taichang, Dayun, and Deda are 181 kg eq.CO₂, 285 kg eq.CO₂ and 330 kg eq.CO₂ respectively during the materials production stage. It is apparent from this figure that material production is the biggest potential CO₂ emission stage in as far as global warming is concerned. In

terms of highway comparison, the global warming potential of Dayun highway was found to be the highest, whilst despite both being asphalt-based, Dayun exhibited greater global warming potential than Taichang highway. The probable reason for this is that the CO₂ emissions, as illustrated in Fig. 3(a), mainly arises from the raw materials production process, which, in the case of Deda highway has more carbon elements in the combined cement and asphaltic materials. By contrast, Taichang and Dayun highways are only associated with asphaltic raw materials.

Based on the materials balance perspective, there are various CO₂ emissions at the materials production stage. However, the three highways have a similar level and proportion of global warming potential at the each stage of the life cycle. The reason for this is that the total carbon source is the same for all the three highways.

Overall, it is apparent from Fig. 3(a) that concrete-based highways, such as Deda, are detrimentally associated with more global warming potential than asphalt-based highways. As theoretically expected, the materials production stage exhibited the greatest threat to global warming potential followed by the transportation stage. For all the three highways evaluated, operation and maintenance were the least detrimental stages to global warming seconded by construction and end-of-life stages for the asphalt-based and concrete-based highways, respectively.

4.2.2. Acidification potential (AP) impacts

As shown in Fig. 3(b), based on the life cycle impact assessment results, a hypothesis can be inferred that Dayun highway, at 50 % regenerated asphalt pavement and 3 cm of the wearing course, is the most environmentally friendly alternative among the three highways evaluated, with the lowest acidification potential (AP) values.

From Fig. 3(b), the total acidification potential values of Taichang highway, Deda highways and Dayun highway are 122.56 kg eq.SO₂, 72.18 kg eq.SO₂ and 96.52 kg eq.SO₂ respectively. These results indicate that Taichang highway has the highest quantitative value of acidification potential indicator and is not an environmentally friendly highway. Based on the relatively low values of Deda highway, Fig. 3(b) also indicates that cement concrete pavements has more potential to reduce acidification than asphalt-based highway pavements.

Compared to the acidification values of the different life cycle stages, the AP values associated with the end-of-life stage are the largest in magnitude, followed by the materials production stage. This means that the end-of-life stage is detrimentally more impactful in terms of highway acidification and is seconded by the materials production stage. On the other hand, the operation and maintenance stage exhibited the least highway acidification potential with the smallest AP values.

4.2.3. Freshwater toxicity (FT) impacts

Water is an essential natural occurring resource, particularly in freshwater [29]. Regrettably, freshwater toxicity (FT) is on an increas-

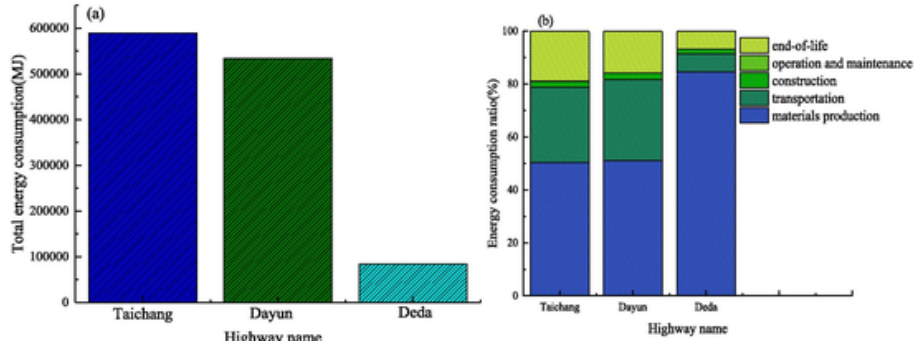


图2. 能源消耗 : (a) 高速公路总能源消耗, (b) 不同生命周期阶段的能源消耗

最大, 而运营和维护阶段的能耗值最小。由于能耗值低, 运营和维护阶段在图2(b)中的数值列几乎不可见。

从图2(b)还可以看出, 材料生产和运输阶段的综合能源消耗比占高速公路整个生命周期的70%以上。这表明运输和材料生产阶段是这三条山西高速公路生命周期中的主要能源消耗阶段。与内蒙古高速公路的能源消耗相比, [28], 生命周期中主要能源消耗阶段不同。然而, 这三条高速公路不同阶段的能源消耗比例不同。与德大公路在材料生产方面的能源消耗相比, 其他两条公路(即大运公路和长太公路)的能源消耗值较小。但是, 所有高速公路在施工阶段的能源消耗似乎相似。

总体而言, 图2(b)的结果表明, 在材料生产阶段, 水泥混凝土高速公路的能耗显著高于沥青混合料高速公路。此外, 还可以得出结论, 材料生产是水泥混凝土高速公路(如德大高速)的主要能耗阶段, 其次是运输和寿命终结阶段。

类似地, 在沥青高速公路中, 材料生产同样被确定为最大的能源消耗者(但数值上低于混凝土高速公路), 随后依次是运输和寿命终结阶段。对于所有三条高速公路, 图2(b)显示施工阶段的能耗相似, 而寿命终结阶段的能耗可忽略不计(即小于1.00%)。与沥青高速公路相比, 水泥混凝土高速公路的材料生产和寿命终结阶段分别对应最高和最低的能耗。

4.2. 生命周期评价环境影响评估

如下文所述, 环境影响评估包括对全球变暖、酸化潜力、淡水毒性和人类毒性的生命周期评价。

4.2.1. 全球变暖潜势 (GWP) 影响

导致全球变暖的因素包括公路施工过程中的卡车运输、原材料生产以及摊铺机排放(CO₂)。因此, 减少CO₂排放是降低全球变暖潜势的关键措施之一。全球变暖潜势(GWP)结果如图3(a)所示。

如图3(a)所示, 在材料生产阶段, 太昌高速、大运高速和德大高速的全球变暖潜势指标分别为181 kg CO₂当量、285 kg CO₂当量和330 kg CO₂当量。从该图可以明显看出, 就全球变暖而言, 材料生产是CO₂排放潜力最大的阶段。在

高速公路比较方面, 大运高速的全球变暖潜势最高; 尽管两者均为沥青路面, 但大运高速的全球变暖潜势高于太昌高速。可能的原因是, 如图3(a)所示, CO₂排放主要来源于原材料生产过程, 而德大高速的沥青与水泥组合材料中含有更多碳元素。相比之下, 太昌高速和大运高速仅使用沥青原材料。

基于物料平衡视角, 材料生产阶段存在多种CO₂排放。然而, 三条高速公路在生命周期各阶段的全球变暖潜势水平和比例相似。原因是三条高速公路的碳源总量相同。

总体而言, 从图3(a)可以明显看出, 以混凝土为主的公路(如德大公路)比沥青公路具有更大的全球变暖潜势。正如理论预期, 材料生产阶段对全球变暖潜势的威胁最大, 其次是运输阶段。对于所评估的三条高速公路, 运营与维护阶段对全球变暖的影响最小, 其次是施工阶段和生命末期阶段(分别针对沥青公路和混凝土公路)。

4.2.2. 酸化潜能影响

如图3(b)所示, 基于生命周期影响评估结果, 可以推断出大运高速公路在50%再生沥青路面和3厘米磨耗层的情况下, 是所评估的三条高速公路中环境友好型最优的方案, 其酸化潜能值最低。

从图3(b)可知, 太长高速公路、得大高速公路和大运高速公路的总酸化潜能值分别为122.56 kg SO₂ eq、72.18 kg SO₂ eq和96.52 kg SO₂ eq。这些结果表明, 太长高速公路的酸化潜能指标量化值最高, 不属于环境友好型高速公路。基于得大高速公路相对较低的数值, 图3(b)还表明, 水泥混凝土路面比沥青路面具有更大的减少酸化的潜力。

与不同生命周期阶段的酸化值相比, 与生命周期结束阶段相关的AP值在数量上最大, 其次是材料生产阶段。这意味着生命周期结束阶段在高速公路酸化方面的影响更为不利, 其次是材料生产阶段。另一方面, 运营和维护阶段显示出最低的高速公路酸化潜力, 具有最小的AP值。

4.2.3. 淡水毒性 (FT) 影响

水是一种重要的天然资源, 尤其是在淡水[29]中。遗憾的是, 淡水毒性(FT)正在上升,

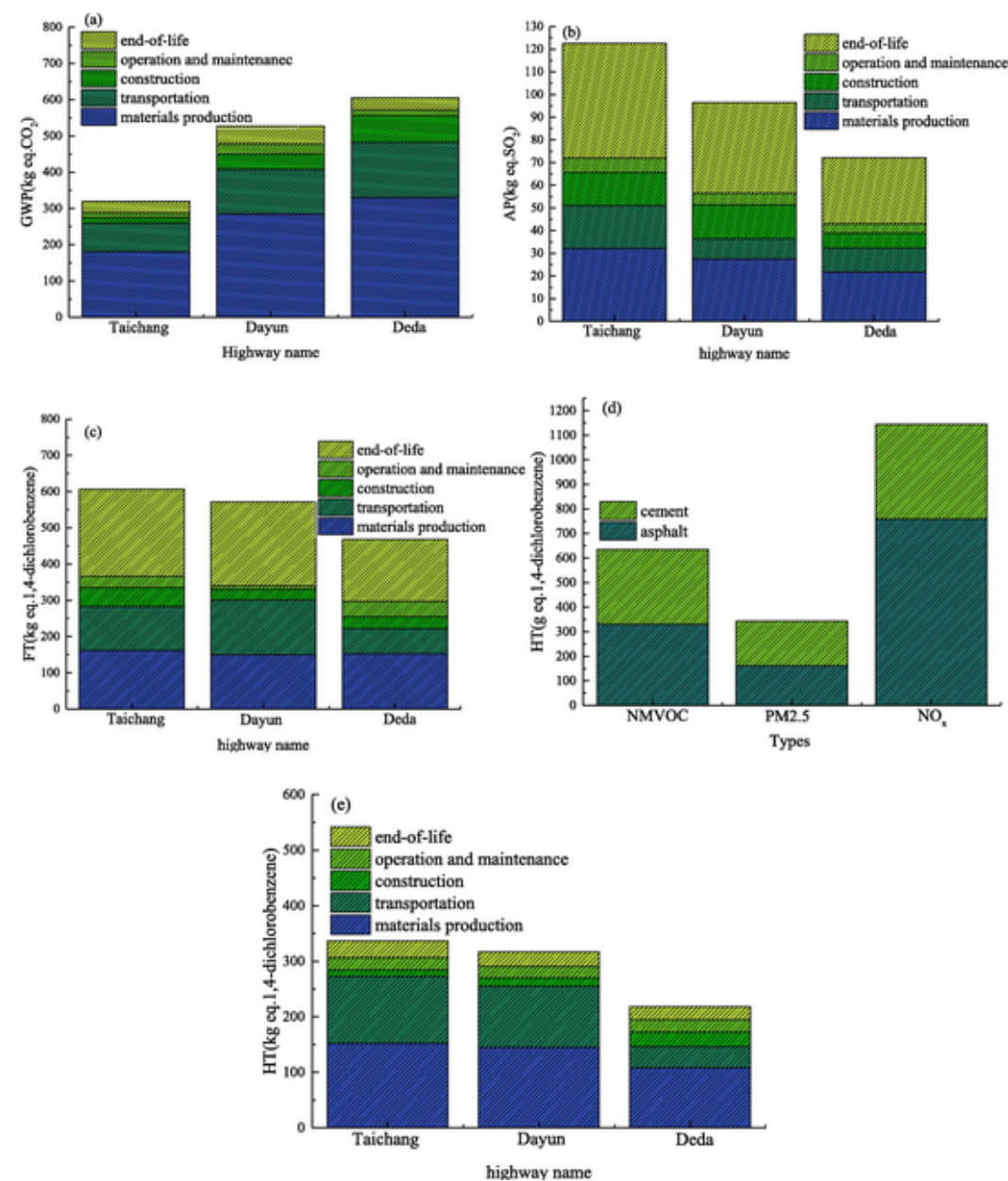


Fig. 3. Environmental pollution (a) highway global warming potential (GWP = Global Warming Potential), (b) highway acidification results (AP = Acidification Potential), (c) Highway freshwater toxicity (FT = Freshwater Toxicity), (d) human toxicity (HT) of materials, (e) human toxicity (HT) of highway comparisons.

ing trajectory and is continuously impacted by the activities of modern civilization including environmental pollutants arising from highways. Under sunlight irradiation, FT converts to an unstable state that is undesirably detrimental to the human health [30]. As illustrated in Fig. 3 (c), Taichang highway has the largest overall freshwater toxicity (FT) values closely followed by Dayun highway, with the end-of-life stage being the most impactful stage for all the highways during their life cycle. This is probably because the highway was built using asphalt mixtures and asphalt mixtures contain heavy metals and lots of hydrocarbons that potentially contribute to polluting the water – and hence, high overall FT values. This also may partly explain why the overall FT values of Taichang and Dayun highways, both asphalt-based pavements, are close to each and not significantly different.

From Fig. 3(c), it can also be observed that end-of-life is the most impactful stage with the highest FT proportional contribution for all the three highways. This is because high-content FT materials are usually wasted, demolished, and/or recycled at the end-of-life stage – thus,

heavily contributing to freshwater pollution. On the other hand, Taichang highway and Dayun highways have almost similar overall FT values but different values for different stages of the life cycle. Although they are both comprised of asphalt-based highways, the slight differences in the pavement structure could be a contributing factor.

Whilst the total FT is relatively lower, the individual FT values for materials production stages and end-of-life stages of Deda highway (Fig. 3(c)) are comparable to those of asphalt-based highways, namely Taichang and Dayun. However, a significant difference exists for the construction stage with the asphalt-based highways being higher than the concrete-based Deda highway. Fig. 3(c) also shows relatively higher FT values for the construction, operation/maintenance, and end-of-life stages for the Taichang highway than Dayun highway and vice versa for the transportation stage. By contrast, the FT impact of materials products appeared to be similar in all the three highways evaluated. Overall, these FT results indicate that the materials production and end-of-life

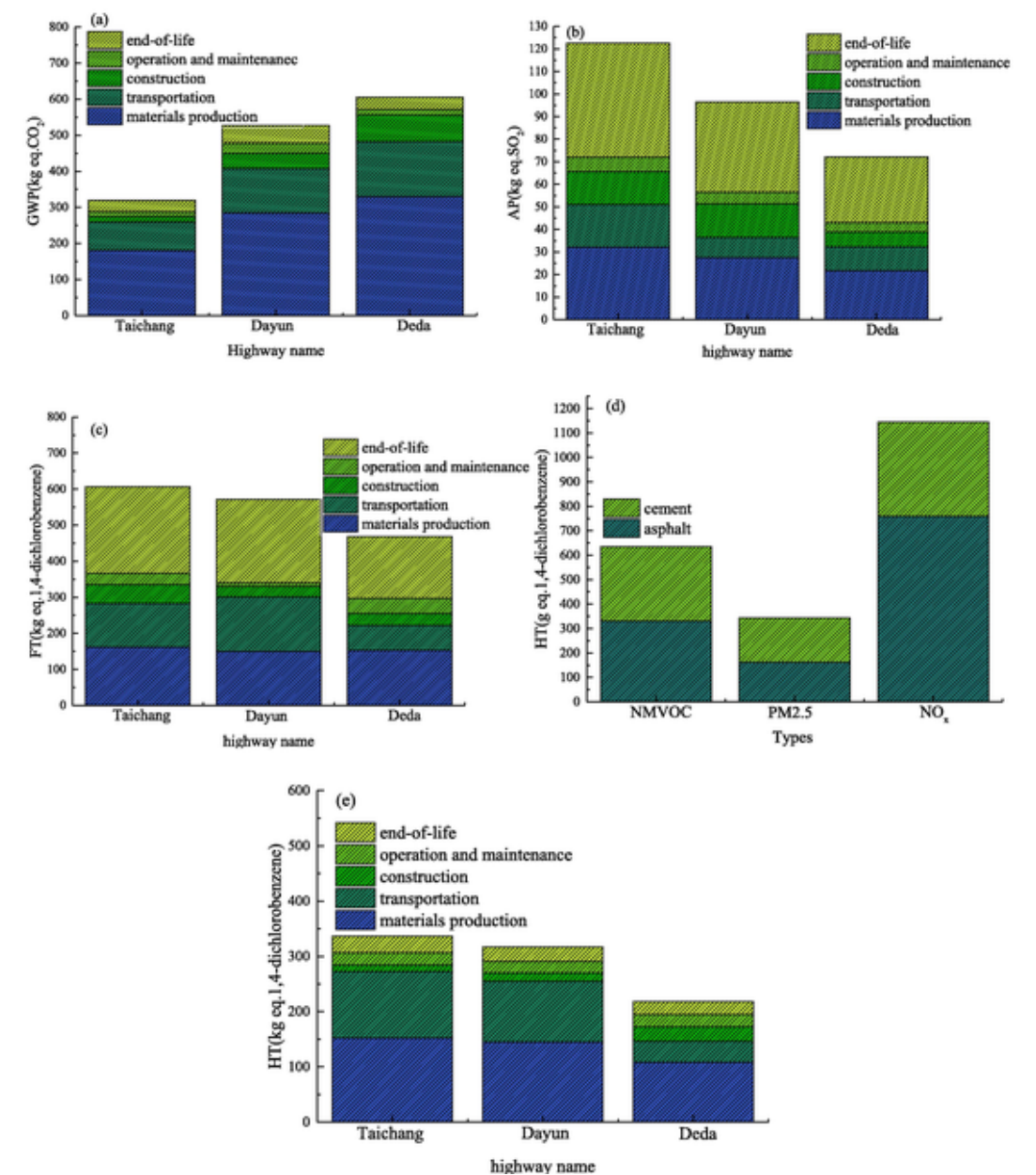


图3. 环境污染：（a）公路全球变暖潜能值（GWP = 全球变暖潜能），（b）公路酸化影响（AP = 酸化潜力），（c）公路淡水毒性（FT = 淡水毒性），（d）材料人体毒性（HT），（e）公路对比的人体毒性（HT）。

并且持续受到现代文明活动的影响，包括高速公路产生的环境污染。在阳光照射下，FT会转化为一种不稳定的状态，对人体健康产生不良影响[30]。如图3(c)所示，太常高速公路的总体淡水毒性（FT）值最大，大运高速公路紧随其后，所有高速公路在其生命周期中，寿命结束阶段的影响最大。这可能是因为该高速公路采用沥青混合料建造，而沥青混合料含有重金属和大量碳氢化合物，容易污染水体，因此总体FT值较高。这也可能部分解释了为什么太常和大运这两条沥青路面高速公路的总体FT值相近且没有显著差异。

从图3(c)还可以看出，生命周期结束阶段是所有三条高速公路中FT比例贡献最高的阶段。这是因为高含量FT材料通常在生命周期结束阶段被废弃、拆除和/或回收—因此，

严重加剧了淡水污染。另一方面，太常高速和大运高速的总FT值几乎相似，但在生命周期不同阶段的值不同。尽管它们都是基于沥青的高速公路，但路面结构的细微差异可能是一个影响因素。

尽管总FT相对较低，但德大高速公路（图3(c)）的材料生产阶段和寿命终结阶段的单个FT值与沥青基高速公路（即太常和大运）相当。然而，施工阶段存在显著差异，沥青基高速公路高于混凝土基的德大高速公路。图3(c)还显示，太常高速公路在施工、运营/维护和寿命终结阶段的FT值高于大运高速公路，而在运输阶段则相反。相比之下，所评估的三条高速公路的材料产品FT影响似乎相似。总体而言，这些FT结果表明材料生产和寿命终结

stages were the dominant sources of freshwater pollution and toxicity, followed by the transportation stage.

4.2.4. Human toxicity (HT) impacts

In this study, the human toxicity (HT) impacts were evaluated using the 1,4-dichlorobenzene equivalent method. 1,4-dichlorobenzene equivalent method used the NO_x, NMVOC, and PM2.5 contents to evaluate the HT impacts. The bigger the NO_x, NMVOC, and PM2.5 contents, the bigger the HT impacts. As shown in Fig. 3(d), the content of NO_x is the highest in magnitude and therefore, most impactful with respect to HT. By contrast, the contents of NMVOC and PM2.5 are comparatively small and therefore, hardly impactful with respect to HT. As shown in Fig. 3(e), the different highways have the different characteristics of HT impacts. Compared to Deda, Dayun and Taichang highways have the comparatively higher HT impacts. The reason for this may be that there are the emissions of VOC and NO_x in the asphalt mixtures-based highways. This is because there is a production of VOC and NO_x when the asphalt is heated above 100 °C. Overall, these results (Fig. 3(d-e)) indicates that compared to cement concrete-based highways, asphalt-based highways have comparatively greater HT impacts. The results (i.e., Fig. 3(d)) further demonstrate that asphalt is more detrimental to the human healthy than cement.

To reduce HT impacts, a recommendation can be made that the government gives more consideration to building the eco-friendly cement concrete-based highways. As shown in Fig. 3(e), the different stages of HT impacts in life cycle shows that the materials production and transportation stages are more impactful and severe than the other stages. Their combined impact proportion is over 85 %. There is a need to be cautious with the materials production and transportation stages during the highway construction process, especially for asphalt-based highways.

Fig. 3(d-e) indicates that there are some HT impacts in the operation and maintenance stage. These emissions mainly happens at the maintenance stage because this stage is associated with regenerating and heating the asphalt. The maintenance stage emits HT gas such as NMVOC and NO_x. The HT at the end-of-life stage is largely associated with the demolishing activities, pavement material removal, truck exhaust emissions, dismantling equipment, etc.

5. Conclusions

In this study, the life cycle assessment (LCA) of three representatives in Shanxi Province, namely Taichang, Dayun, and Deda, was comparatively evaluated. Based on the study results and findings, the following conclusions and recommendations were made:

- Deda highway had the smallest energy consumption indices, with its cement concrete pavement technology rating as a kind of low-energy consumption' pavement technology. The transportation stage and materials production and transportation stages were found to be the primary energy consumers during the life cycle of the highway. The combined energy consumption ratio for these two life cycle stages was over 70 %.
- The global warming potential indicator (Taichang = 181 kg eq.CO₂, Dayun = 285 kg eq.CO₂ and Deda = 330 kg eq.CO₂) had the largest CO₂ emission during the materials production stage. On the other hand, Taichang highway exhibited the greatest potential (AP = 122.56 kg eq.SO₂) for the acidification which suggested that Taichang is not an environmentally friendly highway. Further, the study findings indicated that cement concrete pavement had potential to reduce the acidification potential impact.
- Freshwater toxicity during the materials production stage and end-of-life stages were dominated on Deda highway and Taichang highways, whilst different levels of freshwater toxicity existed on

Dayun highway. Compared to the cement concrete-based highway, asphalt based highways exhibited comparatively higher human toxicity impacts. The materials production and construction stages were found to play a significant role in the human toxicity impacts during the life cycle of highways, with combined HT ratio exceeding 85 %.

- Search for replaced or regenerate materials of petroleum-based asphalt;
- Generalize the new energy-saving paving technology.

CRediT authorship contribution statement

Xinxing Zhou: Writing – original draft, Investigation, Formal analysis. **Xiaorui Zhang:** Writing – review & editing. **Yuan Zhang:** Writing – review & editing. **Sanjeev Adhikari:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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阶段是淡水污染和毒性的主要来源， 其次是运输阶段。

4.2.4. 人体毒性 (HT) 影响

本研究采用1,4-二氯苯当量法评价人体毒性 (HT) 影响。该方法利用NO_x、NMVOC和PM2.5含量来评估HT影响，这些含量越大，HT影响越大。如图3(d)所示，NO_x含量在数值上最高，因此对HT影响最大。相比之下，NMVOC和PM2.5含量较小，故对HT影响甚微。如图3(e)所示，不同公路的HT影响特征各异。与德大公路相比，大运和太昌公路的HT影响较高，原因可能是基于沥青混合料的公路排放了VOC和NO_x。这是因为沥青加热至100°C以上时会产生VOC和NO_x。总体而言，这些结果（图3(d-e)）表明，与水泥混凝土公路相比，沥青公路的HT影响更大。结果（即图3(d)）进一步证明，沥青对人体健康的危害大于水泥。

为了减少人体毒性影响，建议政府更多地考虑建设环保型水泥混凝土公路。如图3(e)所示，生命周期中不同阶段的人体毒性影响表明，材料生产和运输阶段比其他阶段更严重、影响更大。它们的综合影响比例超过85%。在高速公路建设过程中，需要谨慎对待材料生产和运输阶段，尤其是沥青基高速公路。

图3(d-e)表示，在运营和维护阶段存在一些人体毒性影响。这些排放主要发生在维护阶段，因为该阶段涉及沥青的再生和加热。维护阶段会排放NMVOC和NO_x等人体毒性气体。寿命结束阶段的人体毒性主要与拆除活动、路面材料清除、卡车尾气排放、拆解设备等有关。

5. 结论

本研究对山西省三个代表性路段（太昌、大运、德大）进行了生命周期评估（LCA）的对比评价。基于研究结果和发现，得出以下结论与建议：

- 德大高速公路的能耗指数最小，其水泥混凝土路面技术被评为一种“低能耗”路面技术。运输阶段以及材料生产与运输阶段是高速公路生命周期中的主要能源消耗环节。这两个生命周期阶段的综合能耗占比超过70%。

- 全球变暖潜势指标（太昌 = 181 kg eq.CO₂, 大运 = 285 kg eq.CO₂, 德大 = 330 kg eq.CO₂）在材料生产阶段的二氧化碳排放量最大。另一方面，太昌高速公路的酸化潜势（AP = 122.56 kg eq.SO₂）最大，表明太昌并非环境友好型高速公路。此外，研究结果还表明，水泥混凝土路面具有降低酸化潜势影响的潜力。

- 材料生产阶段和寿命末期的淡水毒性在德大高速和太常高速上占主导地位，而不同程度的淡水毒性存在于

大运高速上。与水泥混凝土路面相比，沥青路面表现出较高的人体毒性影响。材料生产和施工阶段在高速公路生命周期中对人体毒性影响起重要作用，合计HT比率超过85%。

- 寻找石油基沥青的替换或再生材料；
- 推广新型节能铺路技术。

CRediT 作者贡献声明

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